

# Data Collection and Preprocessing Phase

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**Project Title:** AI-Enhanced Intrusion Detection System  
**Maximum Marks:** 2 Marks

## Data Quality Report Template

| Data Source                  | Data Quality Issue                                         | Severity | Resolution Plan                                                                                                |
|------------------------------|------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------|
| UNSW-NB15 Dataset (CSV File) | Missing or null values in certain features                 | Moderate | Apply data cleaning techniques such as filling missing values using mean/median or removing incomplete records |
| Network Traffic Features     | Imbalanced dataset (more normal traffic than attack data)  | High     | Use techniques like SMOTE (Synthetic Minority Oversampling) or resampling to balance the dataset               |
| Feature Encoding             | Categorical data not properly encoded                      | Moderate | Convert categorical features into numerical values using Label Encoding or One-Hot Encoding                    |
| Data Normalization           | Features with different scales affecting model performance | High     | Apply normalization or standardization techniques (Min-Max Scaling or StandardScaler)                          |
| Training Data Labels         | Incorrect or inconsistent labeling of attack types         | High     | Validate dataset labels and correct inconsistencies before training                                            |
| Real-Time Input Data         | Incomplete or invalid input provided by user               | Moderate | Implement input validation checks in backend before prediction                                                 |
| Model Input Features         | Missing required features during prediction                | High     | Ensure all required input fields are provided and match training dataset structure                             |
| Dataset Noise                | Presence of irrelevant or noisy data                       | Low      | Apply feature selection and filtering techniques to improve model accuracy                                     |
| Model Overfitting            | Model performs well on training but poorly on new data     | High     | Use cross-validation, regularization, and proper train-test split                                              |
| Data File Handling           | Incorrect file paths or failure to load dataset            | Low      | Use proper file handling methods and verify file paths before execution                                        |

## Summary

The data preprocessing phase plays a crucial role in ensuring the effectiveness of the AI-Enhanced Intrusion Detection System. By identifying and resolving data quality issues such as missing values, imbalance, and improper encoding, the system can achieve higher accuracy and reliability. Proper preprocessing ensures that the machine learning model performs efficiently in detecting and classifying network intrusions in real-time.

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